11.1

Suppose $\{f_n\}$ and f are real valued functions on $\mathbb{R}$, and $f_n \to f$ pointwise, then for any $t \in \mathbb{R}$,

$$\{x \in \mathbb{R} : f(x) \leqslant t\} = \bigcap_{k \geqslant 1} \bigcup_{m \geqslant 1} \bigcap_{n \geqslant m} \{x : f_n(x) < t + 1/k\}.$$

Proof: Note that $\lim_{n\to\infty} f_n(x) = f(x) \leqslant t$ iff for any k exists m, such that for al $n \geqslant m$, $f_n(x) < t + \frac{1}{k}$, which translates to the set relation.

13.1

Suppose $f_1 = f : \mathbb{R} \to \mathbb{R}$ and $f_n(x) = f(f_{n-1}(x))$. If $f_{n_0} = id$ for some n_0 then f is injective.
Proof: If $f(x) = f(y)$, then by induction $f_n(x) = f_n(y)$ so $f_{n_0} = id$ implies $x = y$.

13.2

Prove that there does not exists a continuous function f on $\mathbb{R}$, such that it is a bijection on $\mathbb{Q}^C$, but not on $\mathbb{Q}$.
Proof: Suppose f is bijective on $\mathbb{Q}^C$, $f(\mathbb{R})$ should be connected, hence an interval. Since $\mathbb{Q}^C \subset f(\mathbb{R})$ which is dense in $\mathbb{R}$, $f(\mathbb{R}) = \mathbb{R}$, therefore f is surjective on $\mathbb{R}$.
If $a < b$ and $f(a) = f(b)$, then take $c \in (a, b)$ such that $f(c) \neq f(a)$. Assume $f(c) > f(a)$, then for any $y \in I = (f(a), f(c)) \cap \mathbb{Q}^C$, there exists $u \in (a, c)$ and $v \in (c, b)$ such that $f(u) = f(v) = y \notin \mathbb{Q}$, where at least one of u, v is rational. Hence for each $y \in I$, there exists $u \in (a, b) \cap \mathbb{Q}$ such that $f(u) = y$, but I is uncountable while $\mathbb{Q}$ is countable, a contradiction. Therefore f is bijective on $\mathbb{Q}$.

25.11

Suppose $E \subset \mathbb{R}$ and $|E| < |\mathbb{R}|$, prove that there exists $a \in \mathbb{R}$, such that $E + \{a\} \subset \mathbb{R} \setminus \mathbb{Q}$.
Proof: Note that $\mathbb{Q} - E$ is the image of $\mathbb{Q} \times E$ which has cardinal less than $\mathbb{R}$, hence $\mathbb{Q} - E \subsetneq \mathbb{R}$. Take $a \in \mathbb{R} \setminus (\mathbb{Q} - E)$ then $E + \{a\} \subset \mathbb{R} \setminus \mathbb{Q}$.

25.14

Prove that the set of all transcendental numbers has cardinal $|\mathbb{R}|$.
Proof: Note that the set of algebraic numbers has cardinal $|\mathbb{N}|$: for each n, $A_n = \{x : P(x) = 0 \text{ for some } P \in \mathbb{Z}[x], \deg P = n\}$ is countable since the polynomials of degree n correspond to $\mathbb{Z}^{n+1}$ and each polynomial has at most n roots.

54.7

Suppose $f : [0, 1] \to \mathbb{R}$ and there is some constant M such that for any $x_1, \cdots, x_n \in [0, 1]$ distinct, $|\sum_{k=1}^{n} f(x_k)| \leqslant M$, show that $E = \{x : f(x) \neq 0\}$ is countable.
Proof: Note that $E = \bigcup_n A_n \cup B_n$ where $A_n = \left\{x : f(x) > \frac{1}{n}\right\}$ and $B_n = \left\{x : f(x) < -\frac{1}{n}\right\}$. Then each A_n, B_n is finite, so E is countable.

54.10

Suppose $E \subset \mathbb{R}^2$ is countable, show that there exists disjoint A, B such that $E = A \cup B$, every line parallel to x intersect A at finitely many points, and every line parallel to y intersect B at finitely many points.
Proof: Let U, V be the projection of E to the two axis, which are both countable, we may assume $E = U \times V$. Suppose $U = \{u_n\}$ and $V = \{v_n\}$, let $A = \{(u_n, v_m) : n \leqslant m\}$ and $B = \{(u_n, v_m) : n > m\}$, then they satisfy the requirements.

56.30

Suppose $f : \mathbb{R} \to \mathbb{R}$ is differentiable, and for any t, $\{x : f'(x) = t\}$ is closed. Prove that f' is continuous.
Proof: Note that f' has the intermediate value property. If $g = f'$ is not continuous at some x_0, then there exists n and a sequence $h_k \to 0$ such that $g(x_0 + h_k) \notin B\left(g(x_0), \frac{1}{n}\right)$. WLOG assume $h_k > 0$ and $g(x_0 + h_k) > g(x_0) + \frac{1}{n}$. Take $l_k \in (0, h_k)$ such that $g(x_0 + l_k) = g(x_0) + \frac{1}{2n}$, and WLOG assume $h_{k+1} < l_k$. Then we find a sequence $\{l_k\} \to 0$ with $g(x_0 + l_k) = g(x_0) + \frac{1}{2n}$, so for $t = g(x_0) + \frac{1}{2n}$, $g^{-1}(t)$ is not closed, a contradiction.
Therefore f' is continuous.

57.33

Suppose $f \in C([a, b])$, and non-constant on any subinterval of $[a, b]$. If the critical points of f is dense in $[a, b]$, prove that the set

$$\{x \in [a, b] : f'(x) \text{ does not exist or not continuous}\}$$

is dense in $[a, b]$.
Proof: Otherwise suppose in some subinterval (c, d), f' exists and is continuous. Then the critical points of f in (c, d) are exactly $(f')^{-1}(0)$ which is closed. Hence they are dense implies $f' \equiv 0$ on (c, d), so f is constant in (c, d), a contradiction.

57.41

Suppose F_1, F_2, F_3 are disjoint closed sets in $\mathbb{R}^n$, find $f \in C(\mathbb{R}^n)$ such that $0 \leqslant f \leqslant 1$ and $f|_{F_1} = 0, f|_{F_2} = \frac{1}{2}, f|_{F_3} = 1$.
Solution: Since metrics spaces are normal, by Urysohn's lemma there exists $f_1, f_2 \in C(\mathbb{R}^n)$ such that $0 \leqslant f_i \leqslant 1$, $f_1|_{F_1} = 0, f_1|_{F_2 \cup F_3} = 1$ and $f_2|_{F_1 \cup F_2} = 0, f_2|_{F_3} = 1$. Hence $f = \frac{1}{2}(f_1 + f_2)$ is the desired function.